UNIVERSIDAD DEL ISTMO DE GUATEMALA

FACULTAD DE INGENIERIA

ELECTRONICA 2 Y DISENO ELECTRICO 1

PCIe PI7C9X3G808GP



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NOVIEMBRE DE 2026

UNIS - Facultad de Ingenieria - Electronica 2		
Title: PORTADA		
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INDICE DE HOJAS

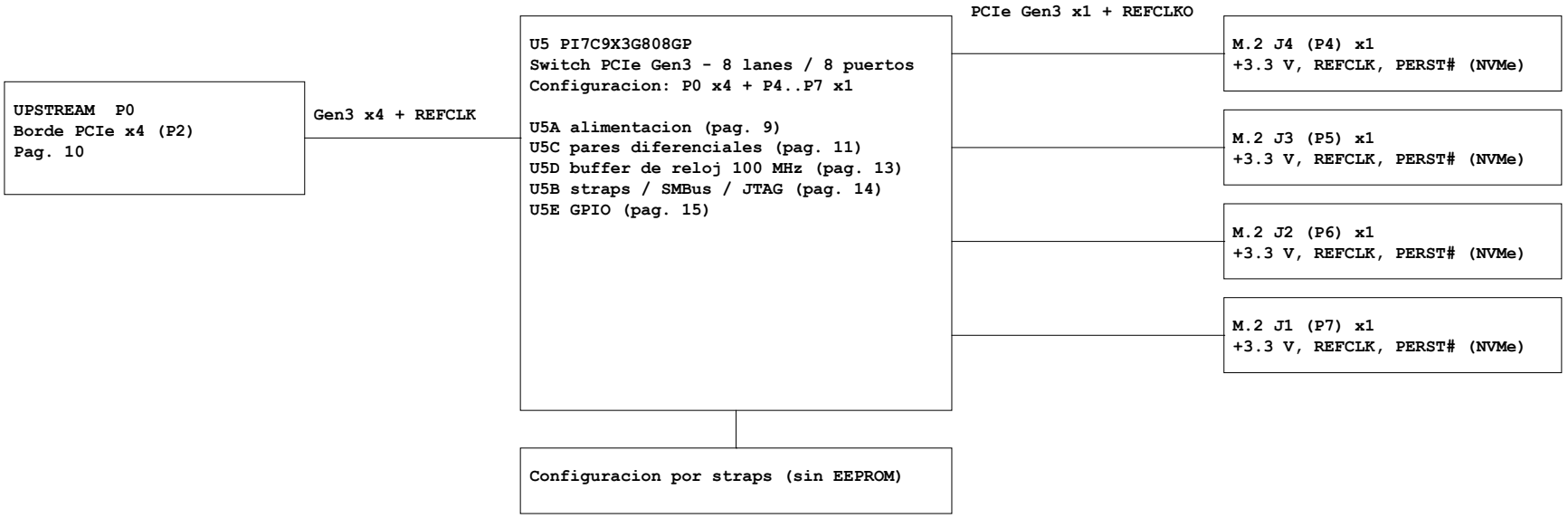
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HISTORIAL DE REVISIONES

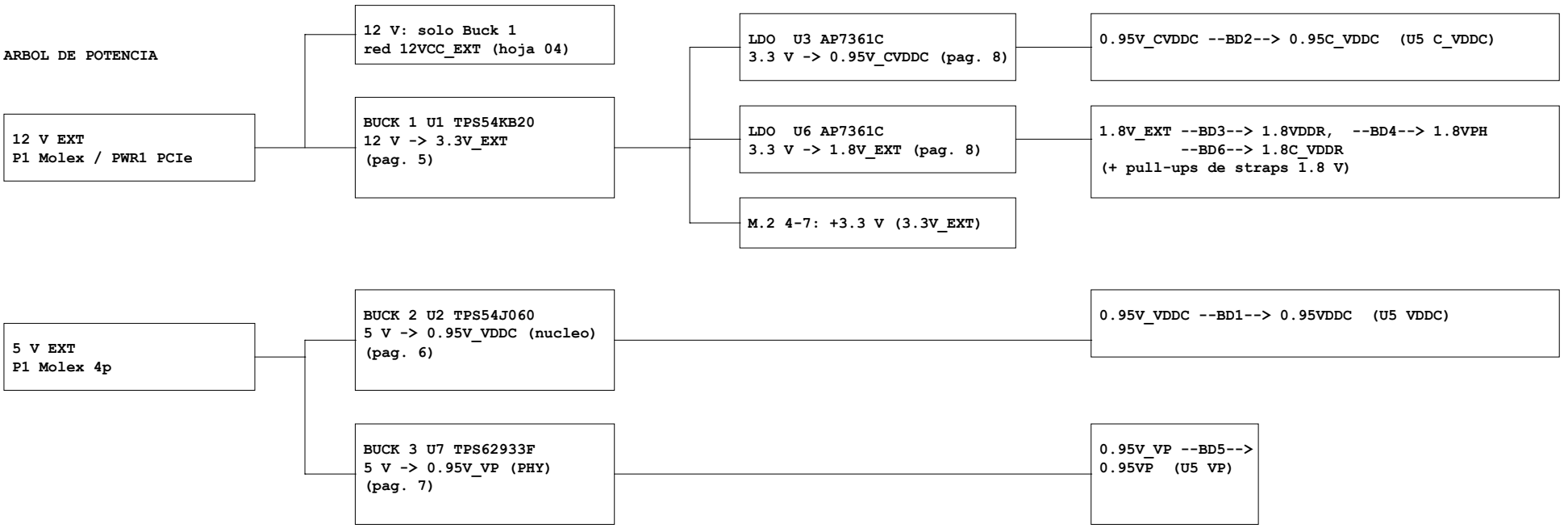
Rev	Fecha	Descripcion
0.1	24/10/2026	Diseno base + 3 convertidores Buck propios (TI WEBENCH)
0.2	28/10/2026	Straps obligatorios y stack-up de 6 capas personalizado
1.0	31/10/2026	Esquematico reorganizado en 15 hojas (potencia y alta velocidad)

DIAGRAMA DE BLOQUES Y ARBOL DE POTENCIA

DIAGRAMA DE BLOQUES (SENALES)



ARBOL DE POTENCIA

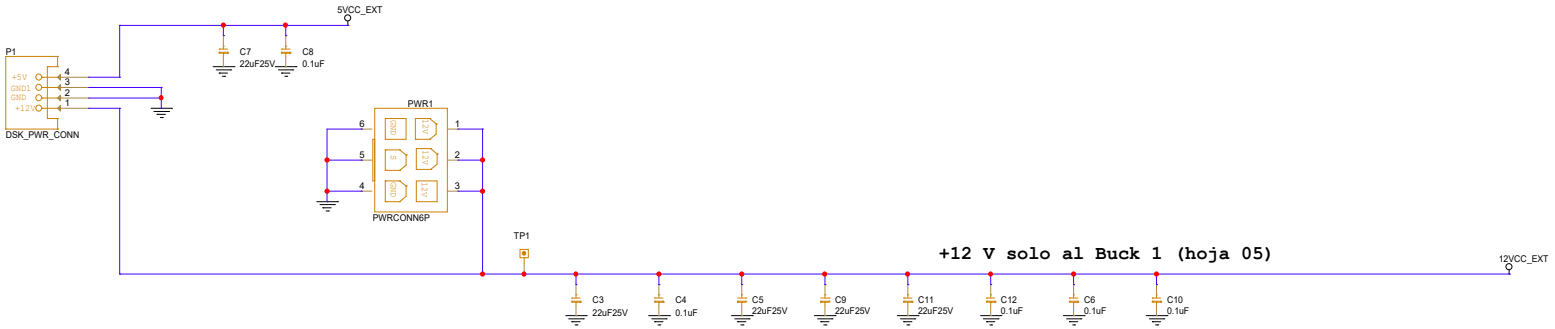


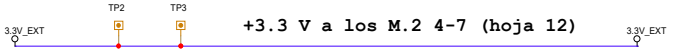
Secuencia de arranque: 3.3V_EXT (PG_3V3) -> EN Buck 2 -> PGOOD (U2_PGOOD) -> EN Buck 3. LEDs indicadores en pag. 4.

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Title: DIAGRAMA DE BLOQUES Y ARBOL DE POTENCIA		
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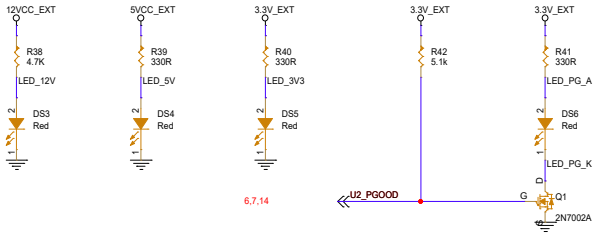
ENTRADA DE POTENCIA 12 V / 5 V

Entrada de 12 V y 5 V externos (P1 Molex 4p / PWR1 PCIe 6p), desacople bulk, alimentacion de los Buck e indicadores LED.





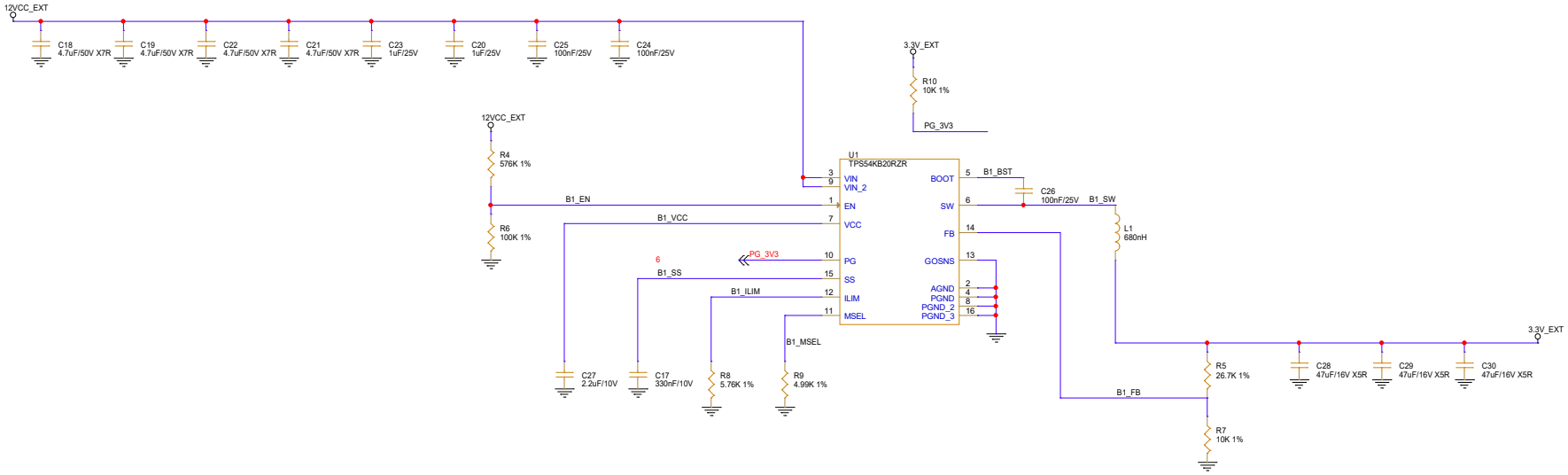
Indicadores: LED por riel y LED de PGOOD de 0.95V_VDDC (Q1)



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Title: ENTRADA DE POTENCIA 12 V / 5 V		
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BUCK 1: 12 V -> 3.3 V

U1 TPS54KB20: 12 V -> 3.3V_EXT. Alimenta los 4 M.2 NVMe (hasta ~2.5 A c/u) y los LDO U3/U6. Su PG (PG_3V3) habilita el Buck 2.



NOTAS DE DISENO - BUCK 1 (TPS54KB20, dimensionado en TI WEBENCH)
Vin = 12 V (10.8 a 13.2 V) Vout = 3.3 V Iout = 15 A fsw = 800 kHz (seleccion por MSEL, R9)
Vout = Vref x (1 + R5/R7) = 0.9 V x (1 + 26.7k/10k) = 3.303 V
UVLO de EN = 9 V (R4/R6) Limite de corriente ~ 21 A (ILIM, R8) Arranque suave ~ 8 ms (C17 = 330 nF)
L1 = 680 nH Coilcraft XAL8080 (37 A) Cout = 3 x 47 uF X5R Eficiencia 94.5 % Rizado 6.8 mVpp
PG (drenador abierto, pull-up R10) -> PG_3V3 = EN del Buck 2 (hoja 6). GOSNS a GND: sensado local.

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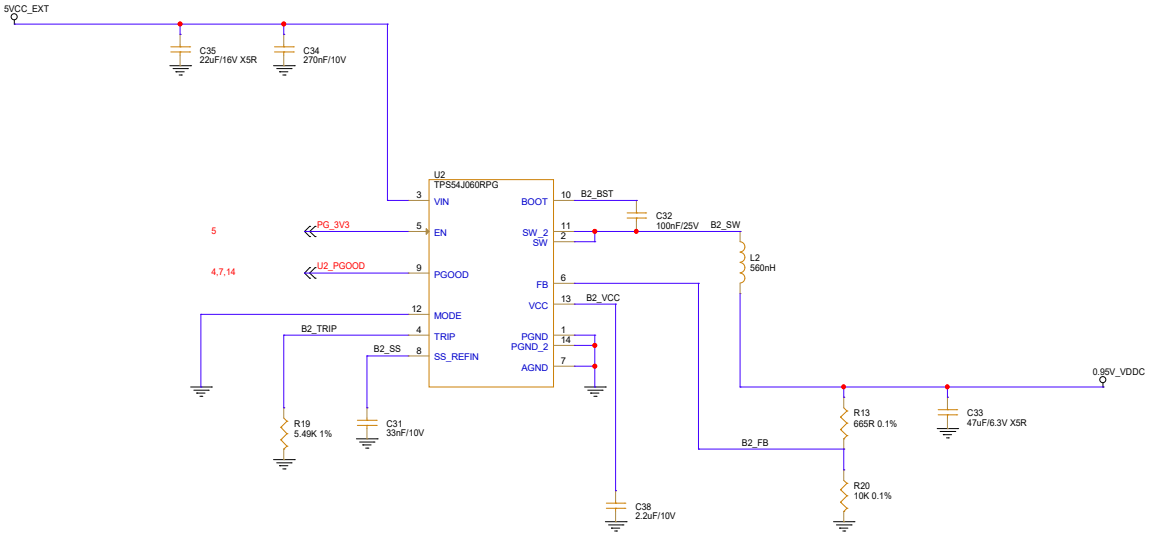
Title: BUCK 1: 12 V -> 3.3 V (TPS54KB20)

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BUCK 2: 5 V -> 0.95 V VDDC

U2 TPS54J060: 5 V -> 0.95V_VDDC (nucleo del switch). EN = PG_3V3. Su PGOOD (U2_PGOOD) habilita el Buck 3 y enciende DS6.

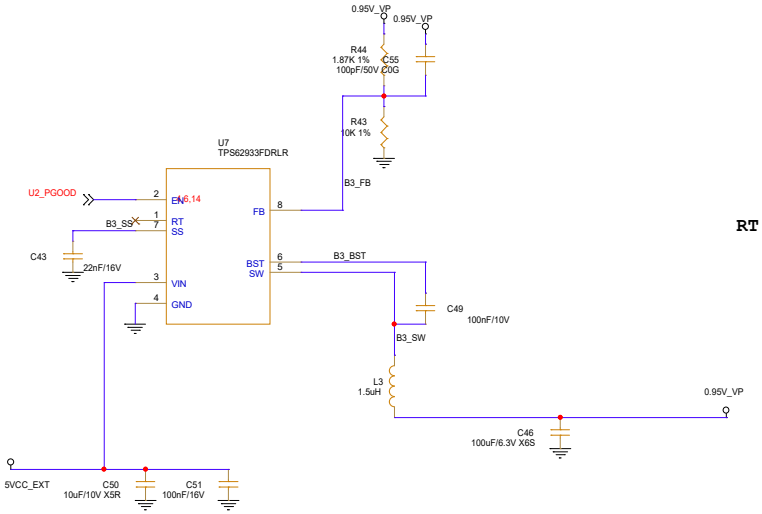


NOTAS DE DISENO - BUCK 2 (TPS54J060, WEBENCH + ajuste propio del setpoint)
Vin = 5 V Vout = 0.96 V Iout = 4 A (VDDC del switch: 3.07 A max) fsw = 1.1 MHz, FCCM (MODE a GND)
Vout = Vref x (1 + R13/R20) = 0.9 V x (1 + 665/10k) = 0.960 V (resistencias de 0.1 %)
Setpoint de 0.96 V: compensa la caida en la ferrita BD1 (31 mV a 3.07 A) -> 0.90 a 0.99 V en las bolas del switch
Limite de corriente ~ 5.5 A (TRIP, R19) Arranque suave ~ 3 ms (C31 = 33 nF) L2 = 560 nH XAL5020 Ef. 84.9 %
EN = PG_3V3 (arranca despues del Buck 1). PGOOD -> U2_PGOOD = EN del Buck 3 (hoja 7) y LED DS6 (hoja 4).

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Title: BUCK 2: 5 V -> 0.95 V VDDC (TPS54J060)		
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BUCK 3: 5 V -> 0.95 V VP

U7 TPS62933F: 5 V -> 0.95V_VP (PHY del switch). EN = U2_PGOOD. Vout = 0.8 V x (1 + R44/R43) = 0.95 V.



RT abierto (NC) = 500 kHz

NOTAS DE DISENO - BUCK 3 (TPS62933F, dimensionado en TI WEBENCH)
Vin = 5 V Vout = 0.95 V Iout = 1 A (VP del switch: 0.57 A max) fsw = 500 kHz, FCCM (PWM forzado)
Vout = Vref x (1 + R44/R43) = 0.8 V x (1 + 1.87k/10k) = 0.950 V C55 = 100 pF de feed-forward
Pin RT abierto = 500 kHz (tabla de RT del datasheet; RT a GND = 1.2 MHz, resistor = 200 kHz a 2.2 MHz)
Arranque suave ~ 3 ms (C43 = 22 nF) L3 = 1.5 uH XAL7030 Cout = 100 uF X6S Ef. 89.4 % Rizado 3.0 mVpp
EN = U2_PGOOD: arranca despues del Buck 2 (secuencia VDDC -> VP).

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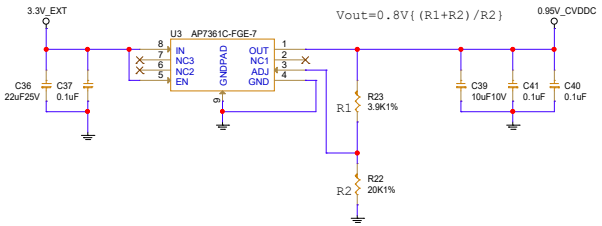
Title: BUCK 3: 5 V -> 0.95 V VP (TPS62933F)

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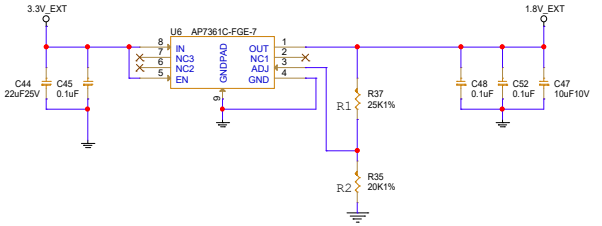
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LDO 0.95 V C_VDDC / 1.8 V

0.95V / 1A

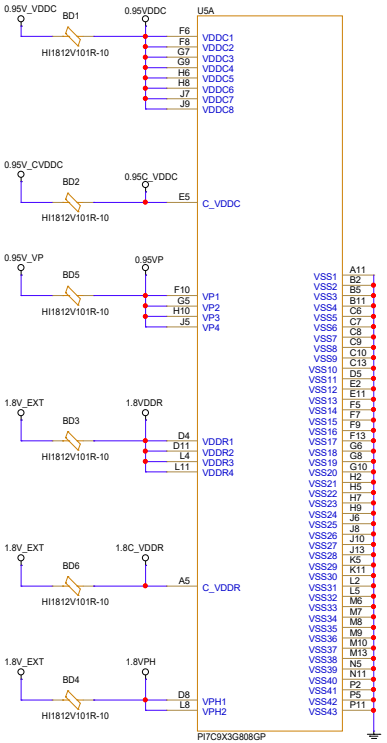


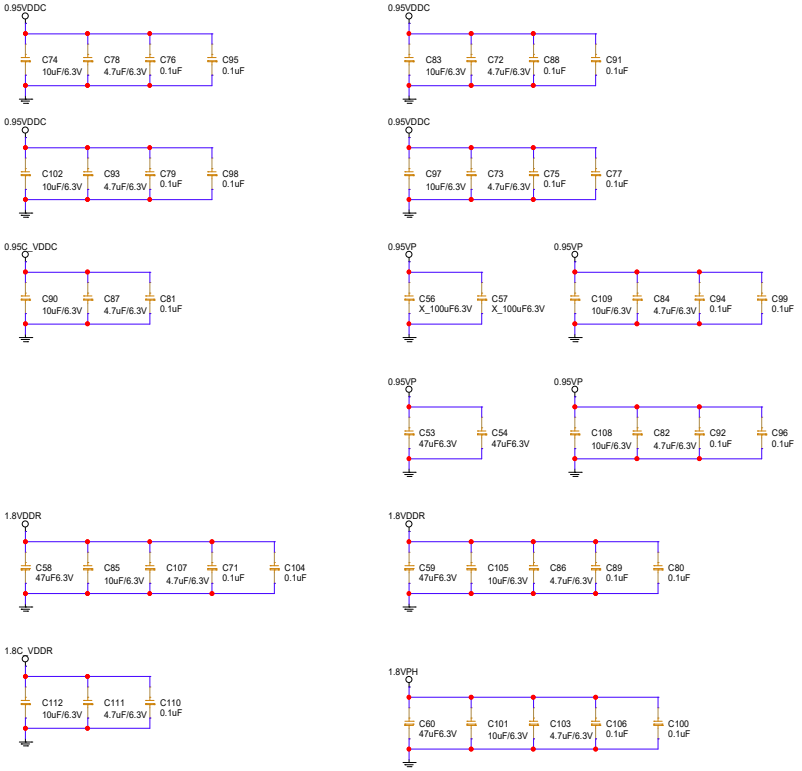
1.8V / 1A



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Title: LDO 0.95 V C_VDDC Y 1.8 V (AP7361C)		
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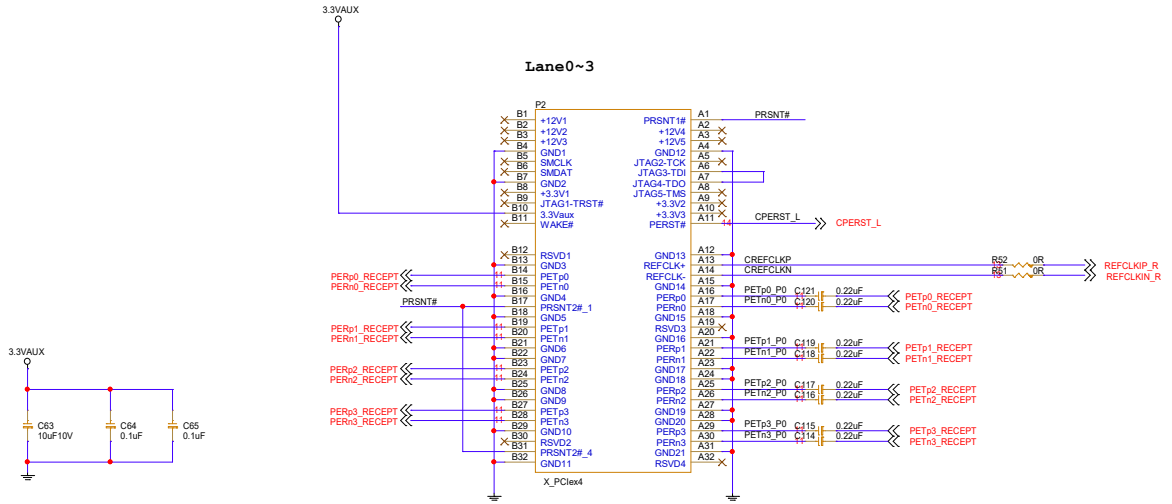
U5A - ALIMENTACION Y DESACOPLE





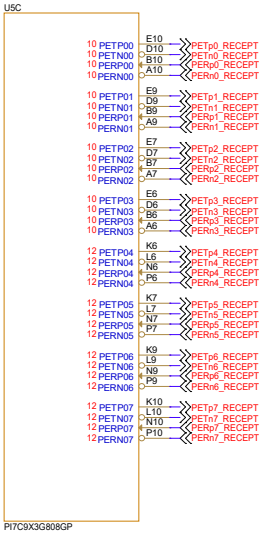
UNIS - Facultad de Ingenieria - Electronica 2		
Title: U5A: ALIMENTACION Y DESACOPLE DEL SWITCH		
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UPSTREAM PCIe x4 (PUERTO 0)



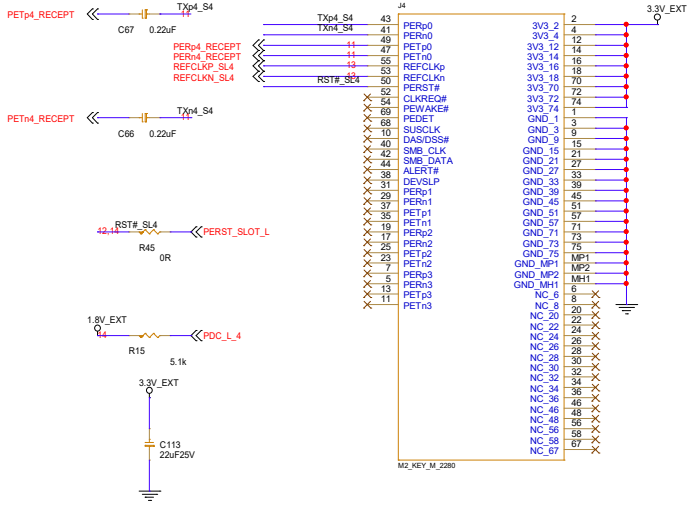
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Title: UPSTREAM PCIe x4 (PUERTO 0)		
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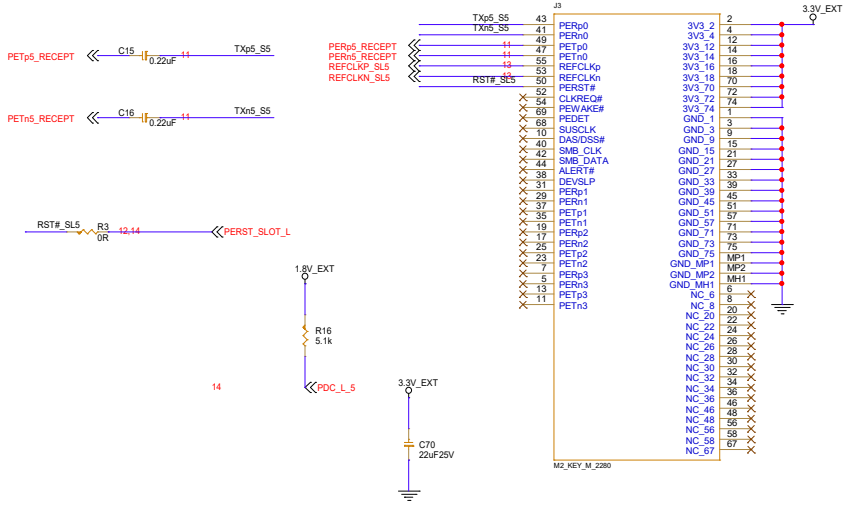
U5C - PARES DIFERENCIALES

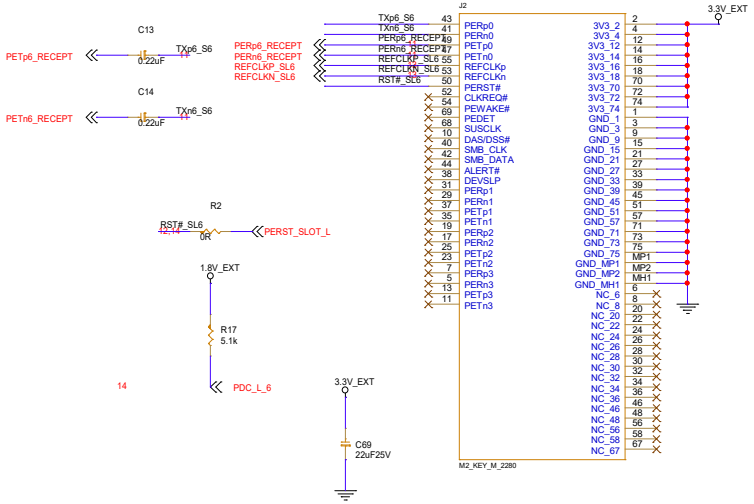


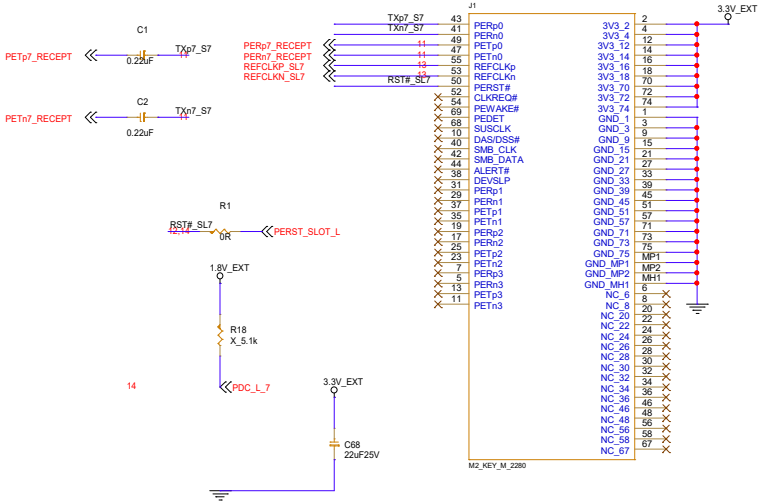
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Title: U5C: PARES DIFERENCIALES PCIe		
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M.2 NVMe - PUERTOS DOWNSTREAM 4 a 7









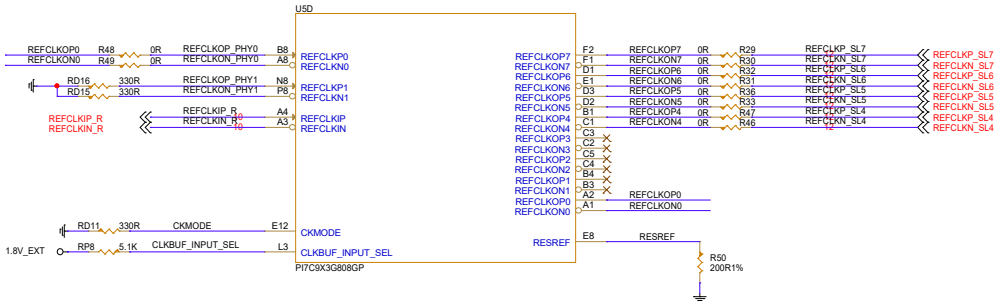
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Title: M.2 NVMe: PUERTOS DOWNSTREAM 4-7

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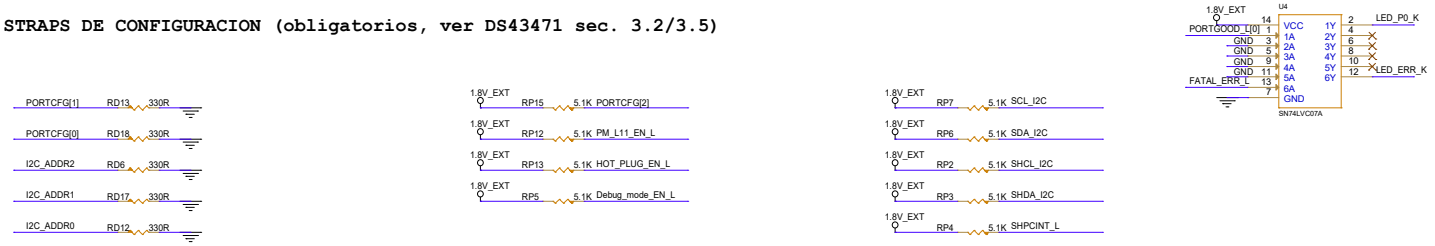
U5D - RELOJ DE REFERENCIA



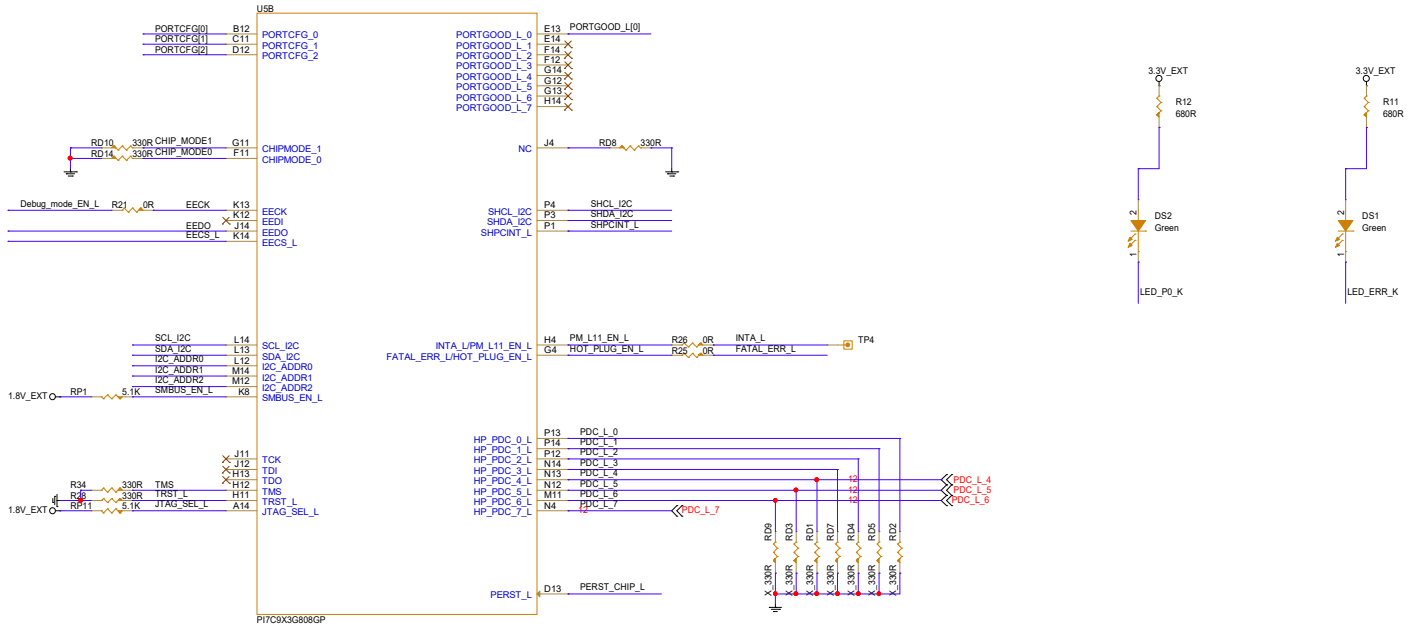
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Title: U5D: RELOJ DE REFERENCIA		
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U5B – CONFIGURACION Y CONTROL

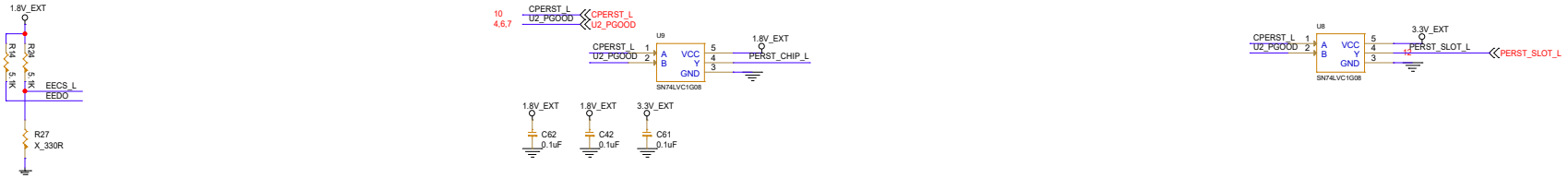
STRAPS DE CONFIGURACION (obligatorios, ver DS43471 sec. 3.2/3.5)



PORTCFG[2:0]=100b: modo 5 puertos (P0 x4 upstream, P4-P7 x1) | I2C_ADDR[2:0]=000 | PM L1.1 y Hot-Plug deshabilitados | Debug mode deshabilitado



PULL-UPS DE LA INTERFAZ EEPROM SPI. R27 = DNP

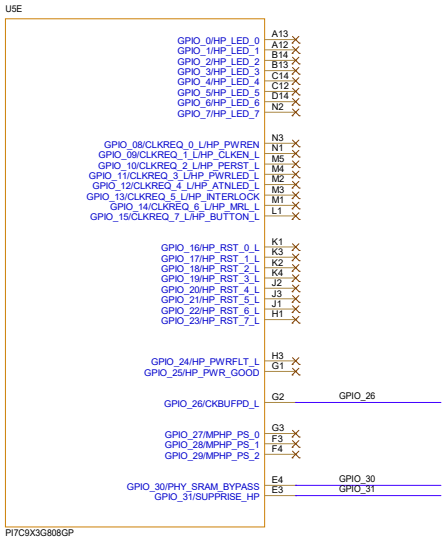


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Title: U5B: CONFIGURACION, STRAPS, SMBus Y JTAG		
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U5E - GPIO

PULL-UPS DE GPIO CON FUNCION ALTERNA: GPIO26=CKBUFPD_L (buffer de reloj activo), GPIO30=PHY_SRAM_BYPASS, GPIO31=SURPRISE_HP





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